

Unidad Técnica de Cálculo

Fecha: 10 01 2024

Tema: Integrales por el método de Fracciones Parciales

Asignatura: Cálculo Integral

$$\int \frac{dx}{x^2 - 5x + 6} = \int \frac{dx}{(x-2)(x-3)} = \frac{A}{(x-2)} + \frac{B}{(x-3)}$$

$$\frac{1}{(x-2)(x-3)} = \frac{A(x-3) + B(x-2)}{(x-2)(x-3)} = \int \frac{dx}{(x-2)} + \int \frac{B}{(x-3)}$$

$$= -\ln(x-2) + \ln(x-3) + C$$

$$x=3$$

$$1 = A(3-3) + B(3-2)$$

$$1 = B \Rightarrow B = 1$$

$$x=2$$

$$1 = A(2-3) + B(2-2)$$

$$1 = -A \Rightarrow A = -1$$

$$\int \frac{x-4}{x^2+8x+12} dx = \int \frac{x-4}{(x+1)(x+2)} dx = \frac{A}{(x+1)} + \frac{B}{(x+2)}$$

$$\frac{x-4}{(x+1)(x+2)} = \frac{A(x+2) + B(x+1)}{(x+1)(x+2)}$$

$$= \int \frac{5}{(x+1)} + \int \frac{6}{(x+2)}$$

$$= 5 \ln(x+1) + 6 \ln(x+2) + C$$

$$x=-2$$

$$-2-4 = A(-2+2) + B(-2+1)$$

$$(-6) = -B \Rightarrow B = 6$$

$$B = 6$$

$$x=-1$$

$$-1-4 = A(-1+2) + B(-1+2)$$

$$-5 = A$$

$$A = -5$$

$$\int \frac{e^{2x}}{e^{2x} + 20} dx = \int \frac{\frac{1}{2} \cdot \frac{dv}{v}}{v^2 + 4v + 20} = \frac{1}{2} \int \frac{dv}{v(v^2 + 4v + 20)} = \frac{1}{2} \int \frac{dv}{v(v+4)(v+5)}$$

$$v = e^{2x}$$

$$v_1 = e^{2x}$$

$$\ln v = \ln e^{2x}$$

$$\ln v = 2x$$

$$x = \frac{\ln v}{2}$$

$$dx = \frac{1}{2} \cdot \frac{dv}{v}$$

$$= \frac{1}{2} \left(\int \frac{\frac{1}{10} \cdot dv}{v} + \int \frac{\frac{1}{4} \cdot dv}{(v+4)} + \int \frac{\frac{1}{10} \cdot dv}{(v+5)} \right)$$

$$= \frac{1}{40} \ln v + \frac{1}{8} \ln(v+4) + \frac{1}{20} \ln(v+5) + C$$

$$= \frac{1}{40} \ln(e^{2x}) + \frac{1}{8} \ln(e^{2x} + 4) + \frac{1}{20} \ln(e^{2x} + 5) + C$$

$$\frac{1}{v(v+4)(v+5)} = \frac{A(v+4)(v+5) + Bv(v+5) + Cv(v+4)}{v(v+4)(v+5)}$$

$$x = 0$$

$$1 = A(0+4)(0+5) + B(0)(0+5) + C(0)(0+4)$$

$$1 = 20A$$

$$\frac{1}{20} = A \Rightarrow A = \frac{1}{20}$$

$$x = -4$$

$$1 = A(-4+4)(-4+5) + 4B(-4+5) + C(-4)(-4+4)$$

$$1 = 4B$$

$$\frac{1}{4} = B \Rightarrow B = \frac{1}{4}$$

$$x = -5$$

$$1 = A(-5+4)(-5+5) - 5B(-5+5) - 5C(-5+4)$$

$$1 = -5C$$

$$\frac{1}{-5} = C \Rightarrow C = -\frac{1}{5}$$

① Derivada

$$2 = 12(x-3)^2 + 6B(x-3) + 2C$$

$$x=3$$

$$2 = 12A(3-3)^2 + 6B(3-3) + 2C$$

$$1 = C$$

$$C=1$$

$$= \int \frac{0 \cdot dx}{(x-3)} + \int \frac{0 \cdot dx}{(x-3)^2} + \int \frac{dx}{(x-3)^3} + \int \frac{80x}{(x-3)^4} + \int \frac{140x}{(x-3)^5}$$

$$= \frac{1}{2(x-3)^2} + \frac{8}{3(x-3)^3} - \frac{14}{4(x-3)^4} + C$$

③ Derivada

$$0 = 24A(x-3) + 6B$$

$$x=3$$

$$0 = 24A(3-3) + 6B$$

$$0 = B$$

$$B=0$$

④ Derivada

$$0 = 24A$$

$$0 = A$$

$$A=0$$

$$\int \frac{x^2 + 5x}{x+5} dx = \int x+5 + \frac{15}{x+5} = \int x dx + \int 5 dx + 15 \int \frac{dx}{x+5}$$

$$5x = x(-3B+C)$$

$$5 = -3(-\frac{1}{3}) + C$$

$$4 = C$$

$$= \int \frac{\frac{4}{3} dx}{(x-3)} - \int \frac{\frac{1}{3}x + 4 dx}{x^2+9} = \frac{4}{3} \ln(x-3) - \frac{1}{3} \int \frac{x dx}{x^2+9} + \int \frac{4 dx}{x^2+9}$$

$$u = x^2+9$$

$$\frac{du}{dx} = 2x$$

$$\frac{1}{(x^2+9)} = \frac{1}{3} \cdot \frac{1}{x^2+3^2}$$

$$= \frac{4}{3} \ln(x-3) - \frac{1}{6} \ln(x^2+9) + \frac{4}{3} \cdot \frac{1}{3} \cdot \frac{x}{3} + C$$

$$\int \frac{dx}{x^3-8} = \int \frac{dx}{(x-2)(x^2+2x+4)} = \frac{dx}{(x-2)(x^2+2x+4)} = \frac{A}{(x-2)} + \frac{Bx+C}{(x^2+2x+4)}$$

$$u = x^2+2x+4$$

$$\frac{du}{dx} = 2x+2$$

$$\frac{1}{2}$$

$$\frac{1}{(x-2)(x^2+2x+4)} = \frac{A(x^2+2x+4) + (Bx+C)(x-2)}{(x-2)(x^2+2x+4)}$$

$$= \int \frac{\frac{1}{12} dx}{(x-2)} + \int \frac{-\frac{1}{2}x dx}{(x^2+2x+4)} + \int \frac{\frac{5}{6} dx}{(x^2+2x+4)}$$

$$= \frac{1}{12} \ln(x-2) - \frac{1}{8} \ln(x^2+2x+4) + \frac{5}{6} \ln(x^2+2x+4) + C$$

$$x=2$$

$$1 = A(2^2+2(2)+4) + (B(2)+C)(2-2)$$

$$1 = 12A$$

$$\frac{1}{12} = A$$

$$1 = \frac{1}{12}(x^2+2x+4) + Bx^2+2Bx+Cx-2C$$

$$x^2 = x^2(\frac{1}{12} + B)$$

$$0 = \frac{1}{12} + B$$

$$-\frac{1}{12} = B$$

$$-\frac{1}{12} = B$$

$$x = x(\frac{1}{6} + 2B + C)$$

$$0 = \frac{1}{6} + 1 + C$$

$$\frac{7}{6} = C$$

$$0 = 2A$$

$$0 = A$$

$$A = 0$$

$$\int \frac{x^2 + 5x}{x-3} dx = \int x+8 + \frac{15}{x-3} = \int x dx + \int 8 dx + 24 \int \frac{1}{x+3}$$

$$\begin{array}{r} x^2 + 5x : x-3 \\ \underline{x^2 + 3x} \\ 8x \\ \underline{-8x + 24} \\ 24 \end{array} \quad = \frac{x^2}{2} + 8x + 24 \int \frac{dx}{x+3}$$

$$= \frac{1}{(x-3)} = \frac{A}{(x-3)} = \int \frac{dx}{(x-3)} = \ln(x-3)$$

$1 = A$

$$= \frac{x^2}{2} + 8x + 24 \ln(x-3) + C$$

$$\int \frac{x^2 + 5x}{(x-3)(x^2+9)} dx = \frac{A}{(x-3)} + \frac{Bx+C}{(x^2+9)}$$

$$\frac{x^2 + 5x}{(x-3)(x^2+9)} = \frac{A(x^2+9) + (Bx+C)(x-3)}{(x-3)(x^2+9)}$$

$$x=3$$

$$(3)^2 + 5(3) = A(9+9) + (B(3)+C)(3-3)$$

$$A = \frac{4}{3}$$

Reinigungs

$$x^2 + 5x = \frac{4}{3}x^2 + 12 + Bx^2 - 3Bx + Cx + 3C$$

$$x^2 = x^2 \left(\frac{4}{3} + B \right)$$

$$1 = \frac{4}{3} + B$$

$$B = -\frac{1}{3}$$